Trade & Ahead: Investment Strategies

UNSUPERVISED LEARNING PRESENTATION

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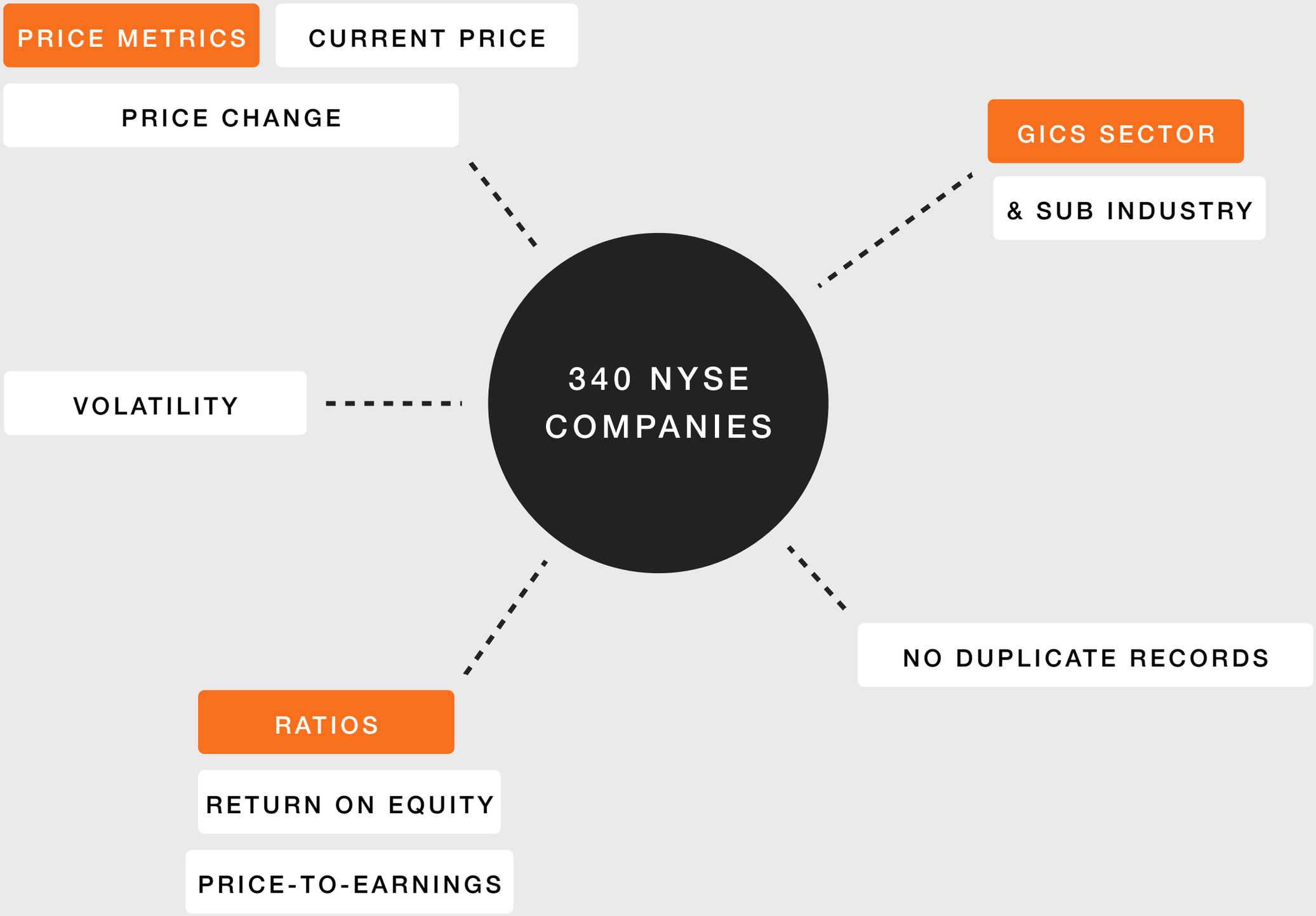
Executive Summary

- K-means clustering was a more useful method for identifying clusters to improve stock analysis
- Hierarchical clustering helped to point out outliers worthy of analysis
- Cluster 0 stocks are not diversified enough
- Cluster 1 should only be invested in if the price of oil is expected to rise
- Invest in cluster 3 for stability
- Invest in cluster 4 for growth

Business Problem Overview

- Predicting trends in the stock market can be overwhelming.
- Diversification is key.
- Clustering will allow us to find stocks with the same characteristics.
- The traits of the clusters can be used to inform investment strategies.
- Investing in a diverse array of stocks which fall into the same cluster can be more predictable than hoping one particular stock will follow some particular trend.

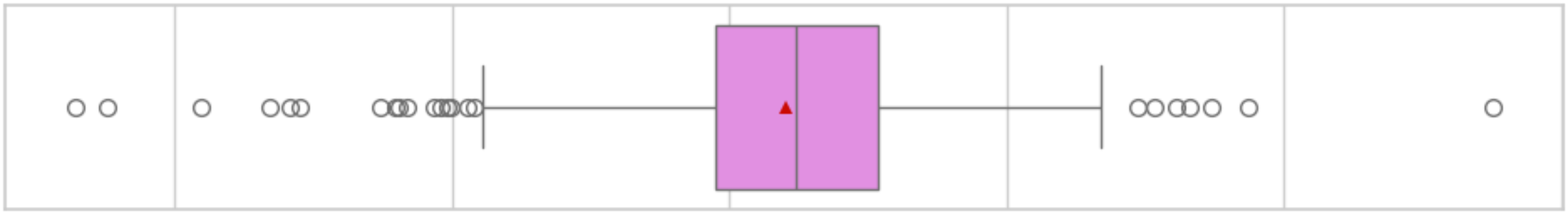
Data Overview

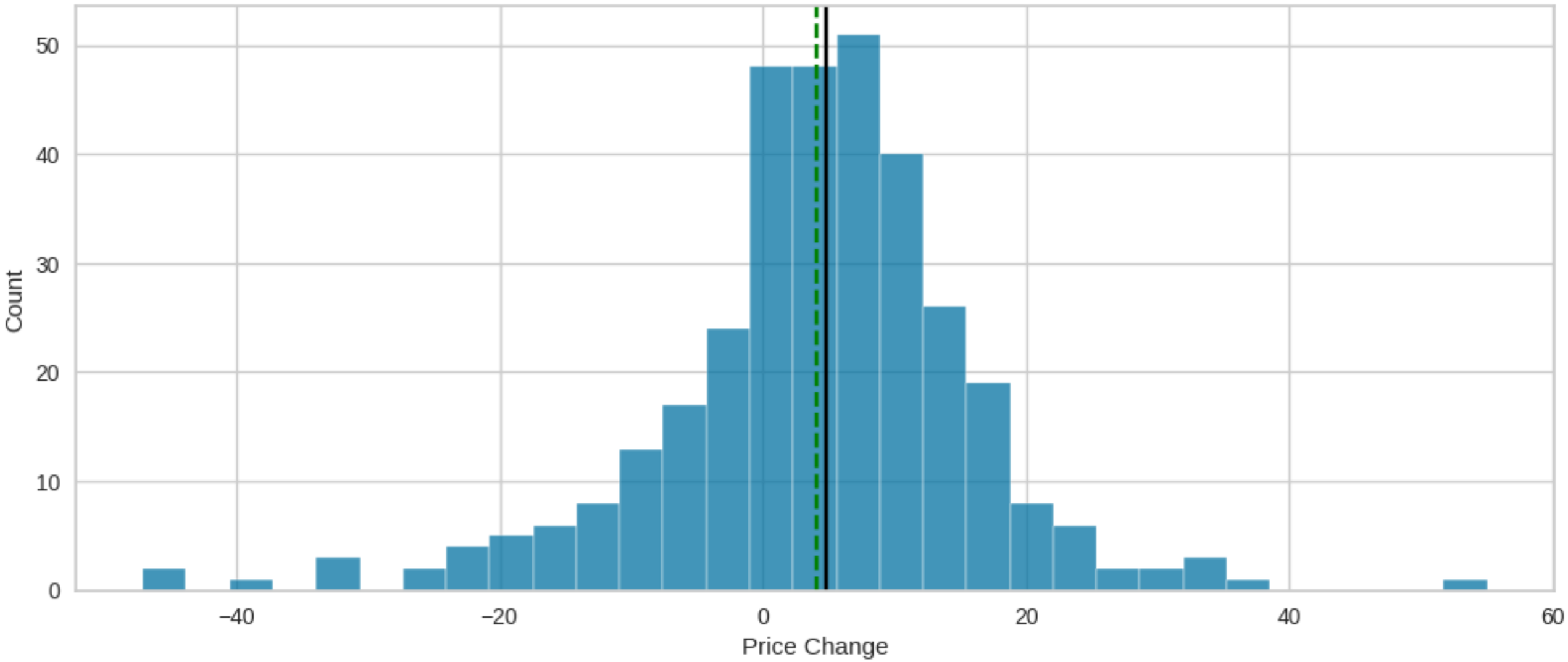


The Price Change Is Normally Distributed.
The Is A Greater Than Zero Average With More Negative Value Outliers.
Diversification Is Key While High Risk Investments Can Be Expensive.

DISTRIBUTION OF

PRICE CHANGE

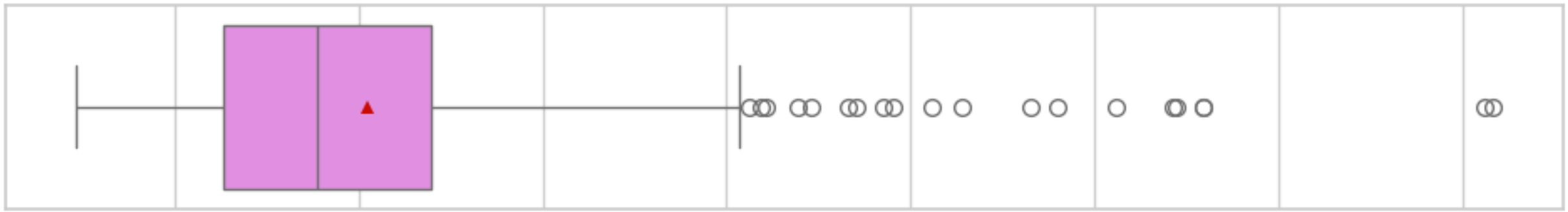


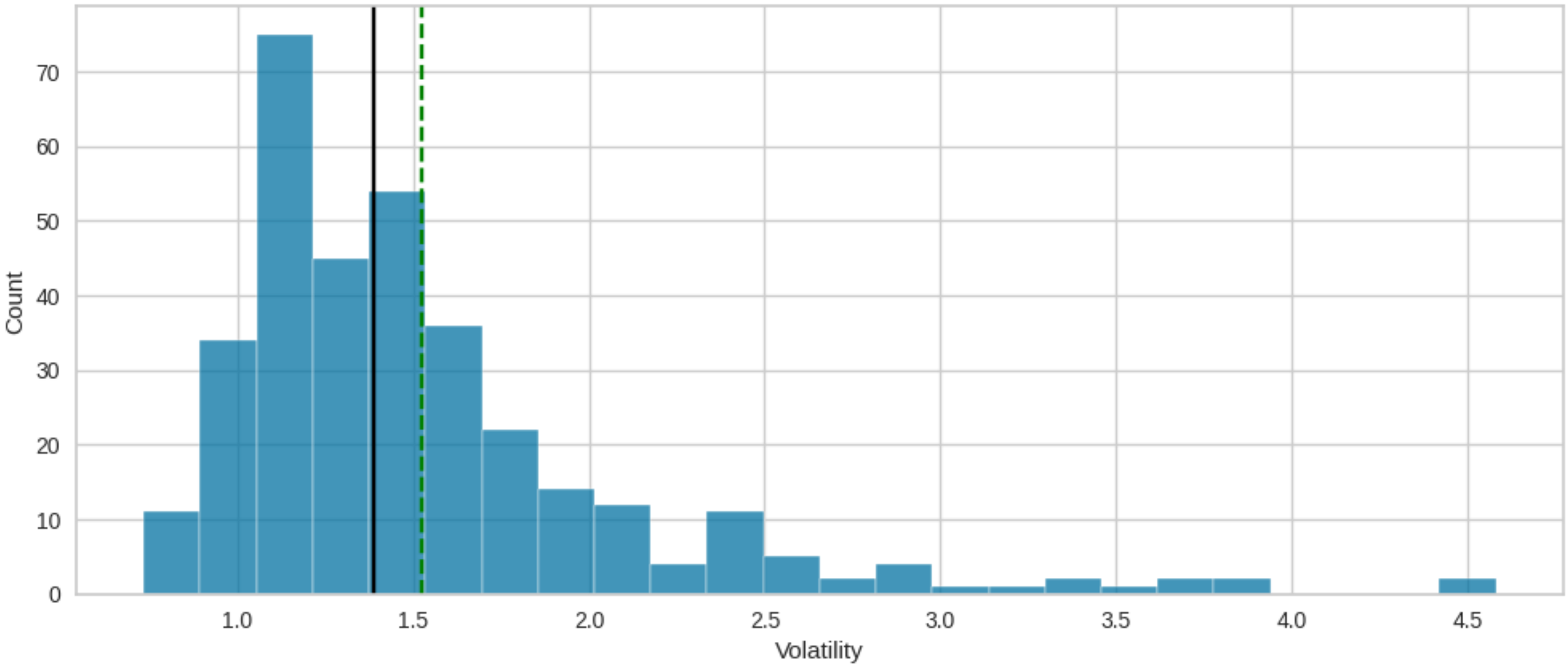


DISTRIBUTION OF

Volatility Is Fairly Consistent With A Few Outliers In The Right Skewed Distribution

VOLATILITY



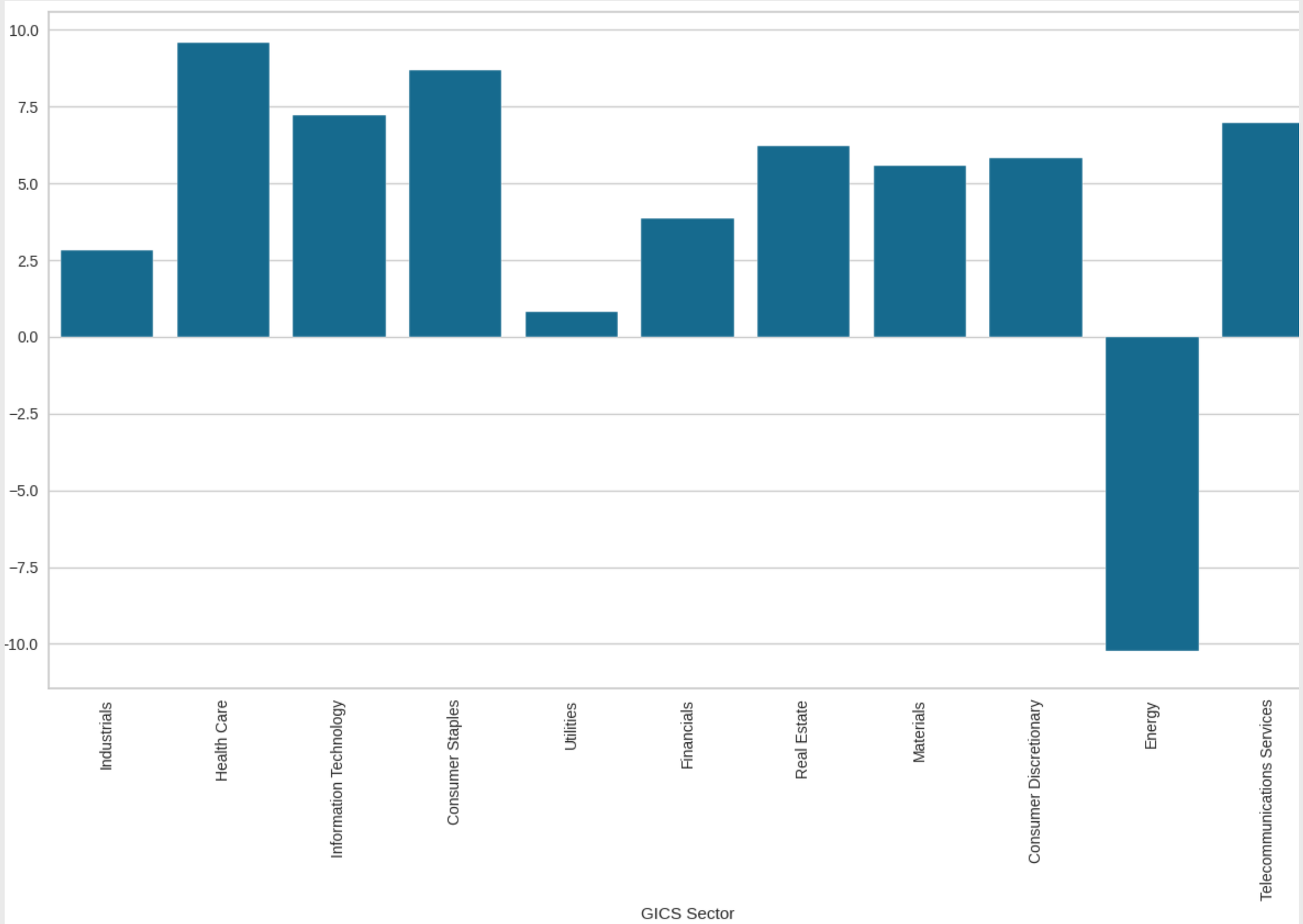


Energy Is A Poor Investment

Health Care And Consumer Staples Have Been Thriving

DISTRIBUTION BY INDUSTRY SECTOR OF

PRICE CHANGE



BIVARIATE ANALYSIS

A Number Of The Correlations Are Implied

Interestingly, Volatility Has A Negative Correlation With Earnings

	Current Price	Price Change	Volatility	ROE	Cash Ratio	Net Cash Flow	Net Income	Earnings Per Share	Estimated Shares Outstanding	P/E Ratio	P/B Ratio
Current Price	1.00	0.13	-0.12	-0.00	0.13	-0.02	0.04	0.48	-0.15	0.26	0.12
Price Change	0.13	1.00	-0.41	-0.04	0.17	0.03	0.18	0.18	0.03	-0.00	0.03
Volatility	-0.12	-0.41	1.00	0.16	0.02	-0.01	-0.38	-0.38	-0.10	0.26	0.05
ROE	-0.00	-0.04	0.16	1.00	-0.07	-0.05	-0.29	-0.41	-0.03	0.02	-0.06
Cash Ratio	0.13	0.17	0.02	-0.07	1.00	0.11	0.03	0.02	0.14	0.09	0.23
Net Cash Flow	-0.02	0.03	-0.01	-0.05	0.11	1.00	0.04	0.02	-0.05	0.03	0.06
Net Income	0.04	0.18	-0.38	-0.29	0.03	0.04	1.00	0.56	0.59	-0.22	-0.06
Earnings Per Share	0.48	0.18	-0.38	-0.41	0.02	0.02	0.56	1.00	-0.03	-0.26	0.01
Estimated Shares Outstanding	-0.15	0.03	-0.10	-0.03	0.14	-0.05	0.59	-0.03	1.00	-0.01	-0.03
P/E Ratio	0.26	-0.00	0.26	0.02	0.09	0.03	-0.22	-0.26	-0.01	1.00	0.04
P/B Ratio	0.12	0.03	0.05	-0.06	0.23	0.06	-0.06	0.01	-0.03	0.04	1.00

Data Preprocessing

PART 1

DUPLICATE & MISSING VALUE CHECK

There were no duplicate values and no missing values; therefore, the data did not need to be treated for such.

There were also no missing values.

PART 2

OUTLIER CHECK

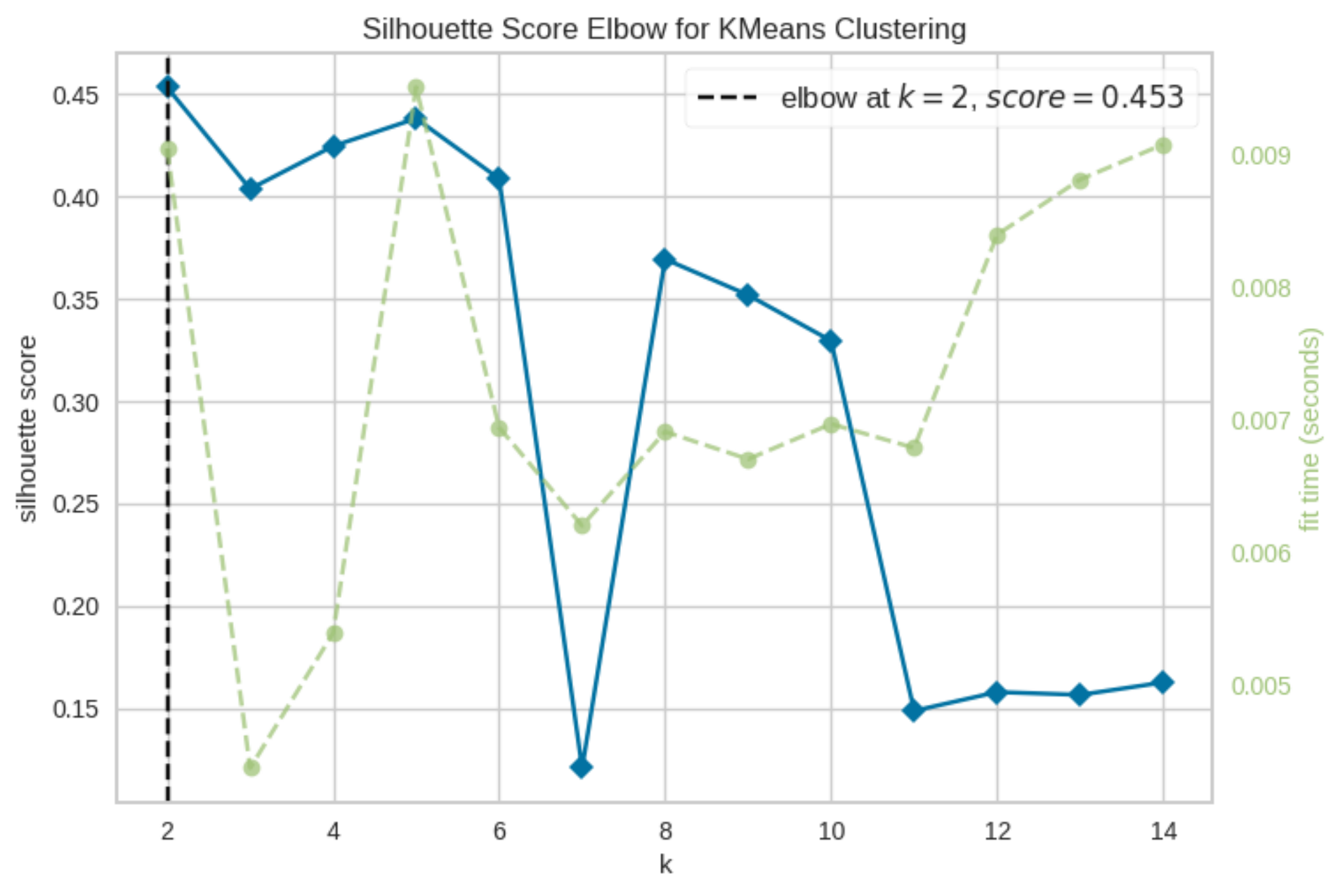
There were outliers in the price to earnings ratios and in the current price, but these likely represented valid data points

PART 3

FEATURE SCALING

A z-score normalization was used to scale the variables.

K = 5 Clusters Will Balance The Need To Distinguish Between Stocks While Providing The Most Utility



Overview Of The Clusters

- Cluster 0 - The majority of stocks, consisting of financials, industrials, and utilities. An average ROE and cash flow. Stable.
- Cluster 1 - A fair number of stocks, mostly consisting of the energy sector. High volatility and dependent on oil prices. Strategic and political.
- Cluster 2 - Small number of stocks, consisting of energy sector outliers. Risky.
- Cluster 3 - Small, consisting of tech and consumption. Not noteworthy.
- Cluster 4 - Small, consisting of healthcare and IT. High price to earnings ratio and price changes. Ideal for growth investments.

Hierarchical Clustering Methods

- Linkage Methods Used: Ward, Complete, Average, and Single
- Average linkage yielded the highest Cophenetic Correlation Coefficient: 0.94
- The resulting dendrogram produced 3 clusters that stood out:
- Cluster 0 had 98% of the data, so the method was not too interesting
- Cluster 1 isolated priceline.com as an outlier
- Cluster 2 isolated Apache and Chesapeake Energy as outliers

Conclusion

- K-means clustering was a more useful method for identifying clusters to improve stock analysis
- Hierarchical clustering helped to point out outliers worthy of analysis
- Cluster 0 stocks are not diversified enough
- Cluster 1 should only be invested in if the price of oil is expected to rise
- Invest in cluster 3 for stability
- Invest in cluster 4 for growth
- Be careful of the outliers exposed by the Hierarchical Clustering analysis